Combinatorial Game Theory

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1 Introduction

Combinatorial Game Theory (CGT) studies two-player games of pure strategy, under perfect information, with no chance, no randomness, and no hidden information.

Imagine the following simple game. Twenty-one tokens are placed on a table. Two players, Rayane and Abdulaziz, take turns removing 1, 2, or 3 tokens. The player who takes the last token wins. At first glance, this appears to be a game of luck. However, a simple but powerful observation determines the outcome under optimal play: if a player begins their turn with a number of tokens congruent to 1 mod 4, they can always force a win by moving to a multiple of 4.

For example, Suppose Rayane begins by removing one token, leaving 20. No matter whether Abdulaziz removes 1, 2, or 3 tokens, Rayane can respond by removing enough tokens to return the total to a multiple of 4. The sequence of positions becomes

$$20,\ 16,\ 12,\ 8,\ 4,\ 0.$$

Therefore, Rayane always takes the last token and wins. What appears to be a playful puzzle hides a precise logical structure. Each configuration can be classified as either a winning or losing position. The key is that mathematics can describe this classification completely. This is the domain of Combinatorial Game Theory (CGT). In classical Game Theory, developed by von Neumann and Morgenstern, we analyze rational decision-making in economics or politics, often involving randomness, hidden information, or probability.

A game (in combinatorial game theory) consists of a set of positions, and from each position a player has a set of legal moves to other positions. Here a set is understood in the usual mathematical

sense, as in number theory: a collection of distinct objects considered as a whole.

A game is called **finite** if every possible sequence of moves eventually terminates; in other words, there is no infinite play. In a finite game, the positions form a finite tree whose leaves are the terminal positions.

A game uses **normal-play** if the player who makes the last legal move wins.

A finite normal-play game is therefore a game that

1. always ends after finitely many moves, and
2. declares the player who makes the final move as the winner.

Because such games always terminate, every position can be classified as either winning (there exists a move to a losing position) or losing (all moves lead to winning positions).

Example. In the 21-token subtraction game, play begins with a pile of 21 tokens, and on each turn a player may remove 1, 2, or 3 tokens. The game must end, since the token count decreases each move, and the player who takes the last token wins. Thus it is a finite normal-play game. We will only focus on finite normal-play games. This brings us to two key theorems that will guide the rest of our work.

In this paper, we will discuss Zermelo's Theorem, which establishes the existence of a determined outcome (existence of a strategy that forces a win, a loss, or a draw) for a broad class of finite games with perfect information. This is a large class of finite games in which each player is aware of all previous moves, and the game is guaranteed to end in a finite number of moves.

Then, we will focus on Sprague–Grundy's theorem, which applies specifically to impartial games, which constitute a subset of finite games with perfect information. While Zermelo ensures the existence of a determined outcome, Sprague–Grundy gives us a quantitative framework to figure out who wins the game and how, but only in the case of impartial games.

2 Basic Terminology and Notation for Combinatorial Game Theory

In this section, we introduce the formal terminology needed to analyze normal-play games rigorously.To reason precisely about games, we will first introduce some formal definitions. We will use the same set notation that we established in class: sets are enclosed in braces , elements are separated by commas, and the vertical bar "—" denotes separation between the possible moves of

the two players.

2.1 Positions, Moves, and Options

Definition 2.1 (Position)**.** A *position* of a combinatorial game is a complete description of the current state of the game together with the player whose turn it is to move. A game G consists of a collection of possible positions, one of which is designated as the current position. From each position, either player may have certain legal moves leading to new positions

Example. In the 21-token subtraction game, a position is given by the number of tokens remaining together with the information of whose turn it is. For example, "12 tokens remaining, Rayane to move" is a position.

Definition 2.2 (Legal Moves and Options)**.** From a position G, the *options* of G are the positions reachable by making a single legal move. We write G^L for the set of options available to the Left player (the player to move), and G^R for the options available to the Right player. In impartial games, $G^L = G^R$.

Example. In the 21-token subtraction game, if it is Rayane's turn and 12 tokens remain, then the options are the positions with 11, 10, or 9 tokens remaining.

Definition 2.3 (Combinatorial Game)**.** A *combinatorial game* satisfies the following:

1. Two players, often called "Left" and "Right" or p_1 and p_2, alternate turns.
2. The game has perfect information: the full state and history are known to both players at all times.
3. There is no chance or randomness (no dice, hidden cards, or probabilistic events).

A position with no legal moves for either player is written $\mathbf{0} = \{ \mid \}$, and is called the Zero Game or Terminal position.

2.2 Impartial and Partizan Games

Definition 2.4 (Impartial Game)**.** A game is *impartial* if the set of legal moves is the same for both players at every position; that is,

$$G^L = G^R.$$

Example. The 21-token subtraction game is impartial, because both players may remove 1, 2, or 3 tokens at every move.

Partizan games are the opposite: they are games in which the available moves may differ between the two players. Examples include chess and checkers, where the sets of legal moves for the two players are distinct.

2.3 Classification of Positions: N-positions and P-positions

We now introduce one of the most fundamental concepts in CGT.

Definition 2.5 (N-position and P-position)**.** In a finite, impartial normal-play game, a position is an *N-position* (Next-player win) if the player whose turn it is to move has a winning strategy. A position is a *P-position* (Previous-player win) if the player whose turn it is to move does not have a winning strategy; equivalently, every legal move leads to an N-position.

Example. In the 21-token subtraction game from the introduction, the initial position (21 tokens with Rayane to move) is an N-position, since Rayane can move to 20 tokens, which is a P-position. Positions with 20, 16, 12, 8, 4, and 0 tokens are P-positions, and positions immediately before them (21, 17, 13, 9, 5, and 1 tokens) are N-positions. Thus in the opening position, Rayane has the winning strategy.

Lemma 2.6 (Characterization of N- and P-positions)**.** In any finite normal-play game:

1. A terminal position is a P-position.
2. A position is an N-position if and only if it has at least one option that is a P-position.
3. A position is a P-position if and only if all of its options are N-positions.

3 Zermelo's Theorem

Theorem 3.1 (Zermelo's Theorem)**.** Consider a game with two players p_1 and p_2. Assume that they both have access to perfect information. Assume that the game is finite. That is, the game tree has a finite number of nodes. Then, p_1 has a winning strategy, or p_2 has a winning strategy, or both p_1 and p_2 have a strategy that would guarantee a draw.

Important remark: Zermelo's theorem focuses on a very broad class of games: finite games with

perfect information. This class includes games that allow draws (partisan games). In that sense, Zermelo does not classify all positions as N or P (for instance, draw positions are neither N nor P). In particular, the N/P classification specifically applies to impartial normal-play games, which cannot admit draws.

3.1 Motivation and relevance of the theorem

One may think that Zermelo's Theorem does not teach us anything new. After all, intuitively, for any such finite game, there are three possible outcomes: a win, a loss, or a draw.

However, the theorem is more subtle than that. Indeed, it reveals something more profound.

What it actually says is that p_1 has a winning strategy regardless of what p_2 does, or p_2 has a winning strategy regardless of what p_1 does, or both players have a way (strategy) to force a draw.

Definition 3.1 (Game Tree)**.** Given a finite normal-play game position g, the **game tree** of g is the directed finite tree constructed as follows:

- Each node represents a game position reachable from g.
- The root node corresponds to g itself.
- There is an edge from node g' to node h if and only if h is an option of g' (that is, if one legal move leads from g' to h).
- Terminal nodes (leaves) are positions with no available moves for either player.

The depth or height of positions g is defined as the maximum number of moves remaining from g until a terminal position is reached.

3.2 Proof (by strong induction on n, number of nodes)

Let $P(n)$ denote the following statement. For any finite game of n nodes with perfect information, the outcome is determined as follows: p_1 has a winning strategy, or p_2 has a winning strategy, or both players have a strategy that guarantees a draw.

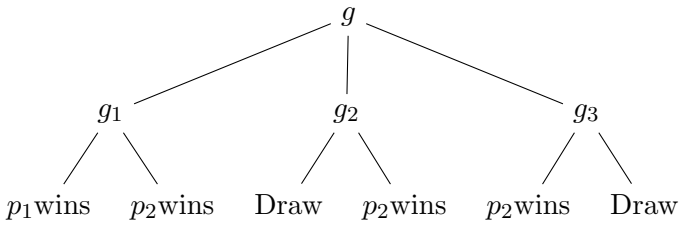
Base Case. Consider a game consisting of a single node. If the game tree has exactly one node, then this node is both the initial position and a terminal position (a state of the game where no legal moves are available). Since p_1 cannot make any move, they certainly cannot make the last legal move. By definition of normal-play games, making the last legal move is the condition to win

such games. Thus the starting player p_1 loses and p_2 wins; in particular, p_2 has a winning strategy (do nothing). Hence $P(1)$ holds.

Inductive Step. Suppose $P(k)$ holds for all integers k such that $1 \leq k \leq n$.

Consider a game of $n+1$ nodes. From the initial position, each legal move (with finitely many possibilities) of p_1 leads to a subgame with at most n nodes. Starting from each of these finite possibilities, we obtain subgames with at most n nodes. In each resulting subgame, p_2 is now the player to move.

In particular, each subgame has a number of nodes $\leq n$. Even though the subgames may have different outcomes, player p_1 doesn't need to use all of them. They choose one subgame (the one with the best outcome available). By the induction hypothesis, there is a strategy for that chosen subgame, which becomes the strategy for the whole game. Then, p_1 has a winning strategy, or p_2 has a winning strategy, or both players have a strategy that guarantees a draw. So $P(n+1)$ is true.



A game tree illustrating the proof. The terminal outcomes are representative only: the argument does not depend on their specific values, but only on the fact that each subgame G_i is determined by the induction hypothesis.

3.3 Example of a partisan game. Tic–Tac–Toe.

The guiding principle is to block all immediate threats while simultaneously creating a double threat whenever possible. At each turn, the player must decide between attacking (creating a potential line of three) and defending (blocking the opponent from creating a line).

Important remark. Note that Tic–Tac–Toe is a partisan game: the available moves depend on which player is moving, since one plays X's and the other plays O's. Since it is a finite game with perfect information, Zermelo's Theorem applies. Here, there is a winning strategy for both players that would end up in a draw. However, this is NOT an impartial game so Sprague-Grundy doesn't apply (it only applies to a subset of finite games with perfect information: impartial games).

Winning strategy by Alyssa Choi:

1. Player 1 places their first move in the center of the square.
2. Player 2 moves anywhere.
3. Player 1 moves so that when Player 2 blocks this path, Player 2 will not be able to form two marks in the same potential winning line (row, column, or diagonal).
4. Player 2 blocks the two marks in the same potential winning line (row, column, or diagonal)
5. Player 1 moves where it will form two possible winning lines (row, column, or diagonal)
6. Player 2 blocks one of the paths.
7. Player 1 wins.

Assuming both players play optimally, the game ends up in a draw.

4 The Sprague-Grundy Theorem

Now that we covered Zermelo's Theorem, we will move on to The Sprague-Grundy Theorem, which is a powerful tool used in CGT to simplify finite, impartial, normal-play games in order to study them more effectively. Firstly, let's introduce the mathematical notion of game addition

4.1 Addition of Games and Positions and Properties

Definition 4.1. Given two games G and H, their disjunctive sum is defined as

$$G + H = \{\, G^L + H,\ G + H^L \mid G^R + H,\ G + H^R\}.$$

In this game, on each turn, a player chooses one component and makes a legal move in that component only.

- A position of $G + H$ consists of a position g of the game G and a position h of the game H. We refer to this position as $g + h$.
- A valid move from $g + h$ is either a valid move in G from g to some other position g' or a valid move in H from h to some other position h'. The resulting position of $G + H$ is then either $g' + h$ or $g + h'$.

- Game positions under the disjunctive sum are commutative and associative, that is, $G+H = H+G$ and $(G+H)+K = G+(H+K)$.

4.2 Mex Operator

An operator which will be later helpful to us when studying the Sprague-Grundy Theorem is the MEX (minimum excluded) operator.

Definition 4.2. Let S be a set of nonnegative integers. MEX(S) is defined to be the smallest nonnegative integer that does not appear in S. (MEX stands for "minimal excluded".)

For instance:

$$\text{MEX}(0,1,2,3,4) = 5, \quad \text{MEX}(0,1,2,3,5) = 4, \quad \text{MEX}(1,2,3,4,5) = 0.$$

4.3 Important Basic Theorems

Now, let's introduce some important theorems that will be later useful for us when proving the Sprague-Grundy Theorem.

Theorem 4.1. For any position g in any finite impartial game, $g+g$ is always a P-position.

Proof: We aim to prove that the second ("previous") player has a winning strategy from $g+g$. Assume Left is at position $g+g$. If Left moves from $g+g$ to $g'+g$, then Right can imitate the same move in the second copy of g, by moving from $g'+g$ to $g'+g'$, and vice versa. So, if Right keeps on imitating Left's moves in the other copy of g, Left will eventually run out of legal moves, so Right will win. □

Theorem 4.2. If g is a P-position, then for any other position h in a finite impartial game, $h+g$ has the same N/P status as h.

Proof: Assume g is a P-position. So, the previous player has a winning strategy against the player who is at position g. We will break down the proof into two cases. First, assume h is a P-position. Then, the previous player always has a winning strategy against the player at position h. Assume, without loss of generality, Left is at position $h+g$. If Left moves to $h'+g$, Right can simply respond with the winning strategy against position h, since h is a P-position. The same logic applies to if conversely Left moves to $h+g'$. Thus, $h+g$ is a P-position.

Now, assume h is an N-position. Then, by Lemma 1.1, it is possibly to move from h to some P-position h'. Then, since g is also a P-position, by Theorem 2.1 $h' + g$ is also a P-position. Since it is possible to move from $h + g$ to the P-position $h' + g$, it must mean that $h + g$ is an N-position, by Lemma 1.1. □

4.4 The Game of Nim

Now, let's introduce the game of Nim, which will be crucial for us. We will introduce the most famous version of the game Nim, which is 3-pile Nim. Two players start with 3 piles (often called *nimheaps*) of stones. On a turn, a player must remove a nonzero amount of stones from any one nimheap. Whoever takes the last remaining stone wins. We denote a single nimheap of size X as X^*. We can analyze this nimheap:

- X^* is an N-position if $X > 0$, since the next player can win in a single move by removing all the stones in the heap.
- X^* is a P-position if $X = 0$

Important remark. Nim is an impartial game: the available moves do not depend on which player is moving. Since it is a finite game with perfect information, Zermelo's Theorem applies. In this setting, there is a winning strategy for one of the players depending on the sizes of the piles: assuming normal play, the first player wins when the piles have different sizes, and the second player wins when the piles have equal size.

4.5 Understanding Game Equivalence in Final Impartial Games

Now, we have to understand what it means for games to be equivalent.

Definition 4.3 (Game Equivalence)**.** Let G_1 and G_2 be two (possibly different) finite, impartial games, and let g_1 and g_2 be positions of G_1 and G_2 respectively. We say that g_1 is **equivalent** to g_2 ($g1 \simeq g2$) if, for any position h of any finite impartial game H, $g1 + h$ has the same N/P status (whether it is an N-position or a P-position) as $g_2 + h$.

In order to check whether two positions are equivalent, just add them up.

Theorem 4.3. *$g_1 \simeq g_2$ if and only if $g_1 + g_2$ is a P-position.*

Proof: Suppose $g_1 \simeq g_2$. By Definition 4.1, $g_1 + g_2$ has the same N/P status as $g_2 + g_2$ (we are treating g_2 as our arbitrary position h in Definition 4.1). By Theorem 2.2, $g_2 + g_2$ is a P-position (since g_2 is a position in a finite impartial game), so $g_1 + g_2$ is also a P-position, since $g_1 \simeq g_2$.

Now, suppose $g_1 + g_2$ is a P-position and let h be any position in the game. By Theorem 2.2, adding a P-position to a game doesn't change its N/P status, which means that $g_1 + h$ has the same N/P status as $(g_1 + h) + (g_1 + g_2)$, since $g_1 + g_2$ is a P-position. Due to associativity and commutativity of game-position addition, $(g_1 + h) + (g_1 + g_2)$ is equivalent to $(g_2 + h) + (g_1 + g_1)$. Once again by Theorem 2.2, $g_1 + g_1$ is a P-position since g_1 is a position in a finite impartial game, so adding it to $(g_2 + h)$ doesn't change its N/P status. Therefore, $(g_1 + h)$ has the same N/P status as $(g_2 + h)$. Thus, by Definition 4.1, $g_1 \simeq g_2$. $\square$

Now, let's apply the definition of game-equivalence to nimheaps.

Theorem 4.4. Two nimheaps are equivalent if and only if they are the same size.

Proof: $\Rightarrow$) The forward direction states that if $X^* \simeq Y^*$, then $X = Y$. The contrapositive of this statement is: if $X \neq Y$, then $X^* \not\simeq Y^*$. Suppose $X \neq Y$. Since $X \neq Y$, it is either the case that $X > Y$ or $X < Y$. Assume, without loss of generality that $Y > X$. If we remove $(Y - X)$ stones from $X^* + Y^*$ we can get the position $X^* + X^*$, which is P-position by Theorem 2.2 Since we can make a move from $X^* + Y^*$ to a P-position, by Lemma 1.1 it must mean that $X^* + Y^*$ is an N-position. However, by Theorem 2.2 we know $Y^* + Y^*$ is a P-position. So, $X^* + Y^*$ and $Y^* + Y^*$ don't have the same N/P status, which means that $X^* \not\simeq Y^*$.

$\Leftarrow$) The backward direction states that if $X = Y$, then $X^* \simeq Y^*$. Suppose $X = Y$. Since the nimheaps have equal sizes, we can treat the positions X^* and Y^* as the same exact positions of two identical games. Then, for any position h of any impartial game H, $X^* + h$ and $Y^* + h$ will have the same N/P status since we are treating X^* and Y^* as identical positions in two separate but identical games. Then, by Definition 4.1, it must mean that $X^* \simeq Y^*$. $\square$

4.6 Sprague-Grundy Theorem and Proof

Now, we are ready to prove the main theorem of our paper, which will explain the vast importance we put into the game of Nim.

Theorem 4.5 (Sprague-Grundy)**.** Any position g in a finite, impartial, normal-play game is equivalent to a nimheap. In other words, there is some nonnegative integer X such that $g \simeq X^*$. We

call X the *nimvalue* of g and write $|g| = X$.

Proof: We will use strong induction on the height h of g, that is on the maximum number of moves from g until a terminal position.

Base Case: $h = 0$. All possible terminal positions have nimvalue 0, so the base case is trivial.

Inductive Step: Assume the hypothesis holds for all possible positions of height less than some h. Now, suppose the position g is of height h. Let $Z = \{g_1, g_2, g_3, ..., g_k\}$ denote all the possible positions you can move to from g. All of these positions in Z have height less than h since the maximum number of moves you can make from g until a terminal position is reached is h, so any position achieved after a move was made from g will have a height less than h. So, the inductive hypothesis applies to all the positions in Z. This means that, all positions $g_i \in Z$ have a nimvalue X_i and are equivalent to some nimheap X_i^* ($g_i \simeq X_i^*$), by the inductive hypothesis.

Earlier, we introduced the MEX operator, which outputs the smallest nonnegative integer that doesn't appear in our input. We established that by the inductive hypothesis, all positions g_i achievable after a move from g will have a nimvalue of X_i. So, we have a set $\{X_1, ..., X_k\}$ of nimvalues of positions in Z. Since Z is the set of positions achievable after a move from g and g has a height of h, there exists a position $g_1 \in Z$ with height $h-1$ which will have some nimvalue $X_j \in \{X_1, ..., X_k\}$. Therefore, g should have a nimvalue of X_j+1. Now, let $Y = MEX(X_1, X_2, ..., X_k)$. By definition of the MEX operator, Y is the greatest nonnegative integer not present in $\{X_1, ..., X_k\}$, so $Y = X_j+1$.

So, the claim we are trying to prove is that $g \simeq Y^*$, that is that g has a nimvalue of Y. To show that $g \simeq Y^*$, by Theorem 4.3 it is enough to show that $g + Y^*$ is a P-position, which by definition of P-position means showing that all positions from $g + Y^*$ lead to an N-position. We will break this down into two cases:

Case 1: Since Z is the set of positions achievable after a move from g, we can make a move from g to some $g_i \in Z$, resulting in the position $g_i + Y^*$. Since g_i will have height smaller than h, g_i will be equivalent to some nimheap X_i^* by our inductive hypothesis. Therefore, the position $g_i + Y^*$ will be equivalent to $X_i^* + Y^*$. By definition of the MEX operator, Y is the greatest nonnegative integer not present in $\{X_1, ..., X_k\}$, and since $X_i \in \{X_1, ..., X_k\}$, we have that $X_i < Y$, so you can move from $X_i^* + Y^*$ to $X_i^* + X_i^*$. By Theorem 4.4, $X_i^* \simeq X_i^*$ since $X_i = X_i$. Then, by Theorem 4,3, $X_i^* + X_i^*$ is a P-position. This in turn means that $X_i^* + Y^*$ was an N-position, which in turn means that $g + Y^*$ was a P-position, since we moved from $g + Y^*$ to $X_i^* + Y^*$, and from $X_i^* + Y^*$ to $X_i^* + X_i^*$. Therefore, since $g + Y^*$ is a P-position, by Theorem 4.3, $g \simeq Y^*$.

Case 2: Conversely, since $Y > X_i$ for all $X_i \in \{X_1, ..., X_k\}$, we can make a move from $g + Y^*$ to some position $g + X_i^*$, by definition of the MEX operator. From $g + X_i^*$ you can move to some $g_i + X_i^*$, where $g_i \in Z$. g_i will have height less than h, so it will be equivalent to some nimheap X_j^* where $X_j \in \{X_1, ..., X_k\}$, by the inductive hypothesis. So, $g_i + X_i^*$ will be equivalent to $X_j^* + X_i^*$. So, from $g + X_i^*$ you can make a move to $g_j + X_i^*$ where g_j will be equivalent to X_i^*. So, from $g + X_i^*$ you can make a move to $X_i^* + X_i^*$, which is a P-position by Theorem 4.3, which means that $g + X_k^*$ was an N-position, which in turn means that $g + Y^*$ was a P-position.

Therefore, we have shown that in both possible cases, $g + Y^*$ is a P-position, which means that $g \simeq Y^*$, by Theorem 4.3. $\square$

The Sprague-Grundy theorem is extremely useful in Combinatorial Game Theory, since now when we analyze finite, impartial games under normal-play, we can just analyze every move as a nimheap, which significantly reduces complexity.

5 Conclusion

In this paper, we first introduced a few basic and foundational principles as well as definitions necessary to achieve the goal of our paper: explain two crucial findings in Combinatorial Game Theory (CGT). In this spirit, we showed how Zermelo's theorem gives us the existence of a determined outcome (existence of a strategy that forces a win, a loss, or a draw) for a broad class of games (finite games with perfect information). Focusing on impartial games, a subset of finite games with perfect information, we discussed Sprague-Grundy's Theorem, which gives us the algebraic tools to figure out who wins and how. In particular, we saw how CGT allows us to decompose complex systems into much simpler games.

Beyond that, CGT is a truly fascinating branch of Mathematics and has many real world applications. In that sense, CGT can be thought of as bridge between abstract mathematics and computational applications. For instance, CGT can be useful to algorithmic design. Indeed, it provides a formal framework and mathematical models (game trees, positions) to rigorously analyze deterministic systems. Furthermore, the theory of Constraint Logic gives us the tools to study the computational complexity of games and puzzles in a structured way.

Work Division: Cécile Chen worked on the Introduction (Section 1) and the majority of Section 2. Rayane Jibre worked on the last part of the Introduction (goals of the paper), Section 3, and the

second part of Subsection 4.3, and Section 5. Abdulaziz Sobirov worked on Section 4 and some of Section 2.

The source we found most interesting and useful was the Crash Course of Combinatorial Game Theory for the Mathcamp 2019 Qualifying Quiz. It formalized many mathematical definitions used in Combinatorial Game Theory, explained the concept of game trees, and introduced us to the Sprague-Grundy Theorem, which all of us found extremely fascinating.

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